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Topic (selected): Singular Value Decomposition for Image Compression on Mars Images

Abstract: This work investigates the application of Singular Value Decomposition (SVD) as a low-rank compression technique for Mars surface images sourced from the HiRISE instrument. Images are processed as raw pixel matrices (loaded from FITS or lossless PNG) with no prior compression applied to the input data. The truncated SVD retains only k dominant singular components, reducing storage by a factor of $mn/[k(m+n+1)]$ while preserving dominant structural features. The best-rank-k approximation property guarantees optimality in the Frobenius norm (Eckart & Young, 1936; Mirsky, 1960). We examine the trade-off between storage (KB/MB) and reconstruction fidelity via relative Frobenius error, PSNR (Wang et al., 2004), and SSIM, and compare SVD against JPEG (Wallace, 1992).

Keywords: Singular Value Decomposition, Low-Rank Matrix Approximation, Image Compression, Truncated SVD, Astronomical Image Processing

1 Introduction

Singular Value Decomposition (SVD) is a fundamental matrix factorisation that decomposes any real matrix $A \in \mathbb{R}^{m \times n}$ of rank r into $A = U\Sigma V^\top$, where U and V are orthogonal and Σ is diagonal with entries $\sigma_1 \geq \cdots \geq \sigma_r > 0$ called the ***singular values*** (Trefethen & Bau, 1997). Retaining only the k largest singular values yields a rank-k approximation A_k that is provably optimal among all rank-k matrices (Eckart & Young, 1936; Mirsky, 1960); this makes SVD the theoretical backbone of low-rank image compression (Brunton & Kutz, 2022). The SVD also provides a numerically robust route to the Moore–Penrose pseudoinverse $A^+ = V\Sigma^+U^\top$ (Penrose, 1955; Golub & Van Loan, 2013). This report unifies both aspects: the theoretical foundations of the SVD and pseudoinverse, and the application of low-rank SVD approximation to Mars surface imagery.

The compression of planetary science images poses unique challenges. Instruments such as the High Resolution Imaging Science Experiment (HiRISE) aboard the Mars Reconnaissance Orbiter produce extremely large datasets; individual images can exceed 2×10^8 pixels and must be transmitted over limited-bandwidth deep-space communication links. Lossy compression that preserves scientifically relevant photometric and astrometric precision is critical. Low-rank SVD approximation, with its optimality guarantee and explicit error control, offers a principled alternative to transform-based codecs such as JPEG. This work investigates whether SVD can deliver acceptable reconstruction fidelity on Mars surface imagery at meaningful compression ratios, and how its storage cost compares with JPEG.

1.1 Historical Development

The SVD originated from a classical problem: can a bilinear form $f(x, y) = x^\top Ay$ be reduced to diagonal shape by two orthogonal coordinate changes? Stewart (1993) documents the 19th-century co-discoverers of $A = U\Sigma V^\top$. Computationally, forming $A^\top A$ squares the condition number and amplifies rounding errors. Golub & Kahan (1965) solved this via Householder bidiagonalisation without ever forming $A^\top A$; Golub & Reinsch (1970) added an implicit shift strategy. Their algorithm remains the basis of LAPACK `DGESVD` (Trefethen & Bau, 1997) and is used throughout this work.

1.2 Preliminaries

Definition 1.1 (Matrix)**.** An $m \times n$ ***matrix*** A is a rectangular array of real numbers with m rows and n columns, written $A = (a_{ij})$ with $a_{ij} \in \mathbb{R}$. Square ($m = n$), tall ($m > n$), and wide ($m < n$) matrices are distinguished (Trefethen & Bau, 1997).

Definition 1.2 (Matrix Rank)**.** The ***rank*** of $A \in \mathbb{R}^{m \times n}$ is the dimension of its column

space $\mathrm{col}(A) = \{Ax : x \in \mathbb{R}^n\}$, which equals the maximum number of linearly independent rows or columns, with $\mathrm{rank}(A) \leq \min(m, n)$ (Golub & Van Loan, 2013).

Definition 1.3 (Orthogonal Matrix)**.** $Q \in \mathbb{R}^{n\times n}$ is ***orthogonal*** if $Q^\top Q = QQ^\top = I_n$, so $Q^{-1} = Q^\top$. Its columns and rows form orthonormal bases; orthogonal matrices preserve Euclidean lengths (Trefethen & Bau, 1997).

Definition 1.4 (Eigenvalue and Eigenvector)**.** Let $A \in \mathbb{R}^{n\times n}$. λ is an ***eigenvalue*** and $v \neq 0$ is the corresponding ***eigenvector*** if $Av = \lambda v$. The multiset of eigenvalues is the ***spectrum*** of A (Trefethen & Bau, 1997).

Definition 1.5 (Symmetric Matrix)**.** $A \in \mathbb{R}^{n\times n}$ is ***symmetric*** if $A = A^\top$. By the Spectral Theorem, every real symmetric matrix is orthogonally diagonalisable as $A = Q\Lambda Q^\top$ (Trefethen & Bau, 1997).

Definition 1.6 (Positive Semi-Definite Matrix)**.** A symmetric matrix A is ***positive semi-definite*** (PSD) if $x^\top Ax \geq 0$ for all x, and ***positive definite*** (PD) if equality holds only for $x = 0$. For any A, $A^\top A$ and $AA^\top$ are always PSD (Golub & Van Loan, 2013).

Definition 1.7 (Singular Values)**.** The ***singular values*** of $A \in \mathbb{R}^{m\times n}$ are $\sigma_i = \sqrt{\lambda_i(A^\top A)}$, where $\lambda_1 \geq \cdots \geq 0$ are eigenvalues of $A^\top A$. If $\mathrm{rank}(A) = r$, then $\sigma_1 \geq \cdots \geq \sigma_r > 0 = \sigma_{r+1} = \cdots$ (Trefethen & Bau, 1997).

Definition 1.8 (Singular Value Decomposition)**.** For any $A \in \mathbb{R}^{m\times n}$ of rank r, the ***singular value decomposition*** (SVD) is $A = U\Sigma V^\top$ with U, V orthogonal and Σ having $\sigma_1 \geq \cdots \geq \sigma_r > 0$ on its diagonal. The SVD exists for every real matrix (Trefethen & Bau, 1997).

Definition 1.9 (Left and Right Singular Vectors)**.** The columns u_i of U are ***left singular vectors*** and columns v_i of V are ***right singular vectors***, satisfying $Av_i = \sigma_i u_i$ and $A^\top u_i = \sigma_i v_i$ (Trefethen & Bau, 1997).

Definition 1.10 (Frobenius Norm)**.** The ***Frobenius norm*** of A is $\|A\|_F = \sqrt{\sum_i \sum_j a_{ij}^2} = \sqrt{\sum_i \sigma_i^2}$, the matrix analogue of the Euclidean norm (Golub & Van Loan, 2013).

Definition 1.11 (Spectral Norm)**.** The ***spectral norm*** of A is $\|A\|_2 = \max_{\|x\|_2=1} \|Ax\|_2 = \sigma_1$, the largest singular value (Trefethen & Bau, 1997).

Definition 1.12 (Condition Number)**.** The ***condition number*** of A is $\kappa(A) = \sigma_1/\sigma_n$. A large κ indicates near-rank-deficiency. Forming $A^\top A$ squares κ, amplifying rounding errors (Golub & Kahan, 1965).

Definition 1.13 (Rank-k Truncated SVD)**.** For $1 \leq k \leq r$, the ***rank-k truncated SVD*** is $A_k = \sum_{i=1}^k \sigma_i u_i v_i^\top$. By Definition 1.16, A_k is the optimal rank-$\leq k$ approximation in Frobenius and spectral norms (Eckart & Young, 1936; Mirsky, 1960).

Definition 1.14 (Low-Rank Approximation)**.** A ***low-rank approximation*** at rank k is $A \approx PQ^\top$ with $P \in \mathbb{R}^{m\times k}$, $Q \in \mathbb{R}^{n\times k}$. The truncated SVD stores $P = U_k\Sigma_k$, $Q = V_k$, requiring $k(m+n+1)$ numbers versus mn (Brunton & Kutz, 2022).

Definition 1.15 (Compression Ratio)**.** The ***compression ratio*** is $\rho(k) = mn/[k(m+n+1)]$. $\rho > 1$ indicates genuine compression (Brunton & Kutz, 2022).

Definition 1.16 (Best Rank-k Approximation (Eckart–Young–Mirsky))**.** Among all B with $\operatorname{rank}(B) \le k$, the truncated SVD A_k minimises:

$$\min_{\operatorname{rank}(B)\le k} \|A - B\|_F = \|A - A_k\|_F = \sqrt{\sum_{i=k+1}^{r} \sigma_i^2},$$

$$\min_{\operatorname{rank}(B)\le k} \|A - B\|_2 = \|A - A_k\|_2 = \sigma_{k+1}.$$

Optimality was established for the Frobenius norm by Eckart & Young (1936) and extended by Mirsky (1960).

Definition 1.17 (Relative Reconstruction Error)**.** The ***relative reconstruction error*** is $\varepsilon_k = \|A - A_k\|_F / \|A\|_F = \left(\sum_{i>k}\sigma_i^2 / \sum_{i=1}^{r}\sigma_i^2\right)^{1/2}$ (Trefethen & Bau, 1997).

Definition 1.18 (Singular Value Decay)**.** ***Singular value decay*** is the rate at which σ_i decreases. Smooth images exhibit rapid decay (geometric or power-law), concentrating energy in few components; noisy or textured images decay slowly (Brunton & Kutz, 2022).

Definition 1.19 (Pixel Matrix)**.** A greyscale image of height m, width n is a ***pixel matrix*** $A \in \{0,\dots,255\}^{m\times n}$. Colour images are triples (A_R, A_G, A_B); SVD is applied per channel (Andrews & Patterson, 1976).

Definition 1.20 (Householder Reflection)**.** A ***Householder reflection*** $H = I - 2vv^\top/\|v\|_2^2$ reflects vectors across the hyperplane orthogonal to v, zeroing sub-diagonal entries without forming $A^\top A$, preserving κ (Golub & Kahan, 1965).

Definition 1.21 (Bidiagonal Reduction)**.** The ***Golub–Kahan bidiagonalisation*** (Golub & Kahan, 1965; Golub & Reinsch, 1970) uses Householder reflections to reduce A to bidiagonal form $B = U_L^\top A V_R$, then applies QR iteration to B, avoiding squaring $\kappa(A)$.

Definition 1.22 (Peak Signal-to-Noise Ratio)**.** PSNR of reconstructed $\hat{A}$ relative to original A (pixel range $[0, 255]$) is

$$\mathrm{PSNR}(\hat{A}) = 10\log_{10}\left(\frac{255^2}{\mathrm{MSE}}\right)\ [\mathrm{dB}],$$

where $\mathrm{MSE} = \|A - \hat{A}\|_F^2/(mn)$. Higher PSNR indicates better fidelity (Wang et al., 2004).

Definition 1.23 (Structural Similarity Index)**.** The ***structural similarity index*** (SSIM)

(Wang et al., 2004) compares luminance, contrast, and structure on local patches, yielding a score in $[-1, 1]$ with SSIM $= 1$ for identical images. SSIM correlates more closely with human perception than PSNR.

Definition 1.24 (Photometric Precision)**.** ***Photometric precision*** is the accuracy with which pixel intensities (proportional to photon flux) are preserved after compression (Pence et al., 2010).

Definition 1.25 (Astrometric Precision)**.** ***Astrometric precision*** is the accuracy of source positions after compression. Artefacts distorting pixel distributions shift computed centroids (Pence et al., 2010).

Definition 1.26 (Moore–Penrose Pseudoinverse)**.** For $A \in \mathbb{R}^{m\times n}$ with SVD $A = U\Sigma V^\top$ and rank r, the ***Moore–Penrose pseudoinverse*** is $A^+ = V\Sigma^+U^\top$, where Σ^+ transposes Σ and replaces each non-zero σ_i with $1/\sigma_i$ (Penrose, 1955; Golub & Van Loan, 2013; Trefethen & Bau, 1997).

2 Mathematical Results

2.1 Functional Computation of the SVD via the Eigenvector Method

We detail the algebraic procedure for computing $A = U\Sigma V^\top$ via eigendecompositions of the Gram matrices; for numerical work, Golub–Kahan bidiagonalisation (Definition 1.21) is preferred as it avoids squaring κ (Golub & Kahan, 1965).

Step 1: Calculate the right singular matrix V: Solve $(A^\top A)v_i = \lambda_i v_i$ with $\lambda_1 \geq \cdots \geq \lambda_n \geq 0$. Orthonormal eigenvectors v_i form the columns of V (Trefethen & Bau, 1997).

Step 2: Calculate the left singular matrix U: Solve $(AA^\top)u_i = \lambda_i u_i$. Non-zero eigenvalues of $AA^\top$ match those of $A^\top A$ (Trefethen & Bau, 1997). Eigenvectors u_i form U.

Step 3: Calculate the singular values and construct Σ: $\sigma_i = \sqrt{\lambda_i}$ for $i = 1, \ldots, r = \operatorname{rank}(A)$ fill the diagonal of Σ.

Step 4: Verify the signs of the singular vectors: Verify $Av_i = \sigma_i u_i$ for $i = 1, \ldots, r$. If $Av_i = -\sigma_i u_i$, flip the sign of u_i (or v_i) to restore consistency (Golub & Van Loan, 2013).

Step 5: Verify the reconstruction and calculate error checks: For an exact SVD, the Frobenius norm of the residual must vanish: $\|A - U\Sigma V^\top\|_F = 0$. In finite-precision arithmetic, this residual should be at or below machine epsilon times $\|A\|_F$.

Truncation error (Eckart–Young–Mirsky bound): When the rank-k approximation A_k is used, the Eckart–Young–Mirsky theorem (Definition 1.16) gives the exact Frobenius-norm truncation error:

$$\|A - A_k\|_F = \sqrt{\sum_{i=k+1}^{r} \sigma_i^2}\,, \qquad \text{(Eckart \& Young, 1936).}$$

This formula can be evaluated directly from the singular values without recomputing A_k, making it a cheap diagnostic for choosing k.

Example 2.1 (SVD of a 3×2 matrix)**.** Let

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

We seek a singular value decomposition

$$A = U\Sigma V^\top.$$

First, compute

$$A^\top A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

The characteristic polynomial is

$$\det(A^\top A - \lambda I) = \det \begin{pmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{pmatrix} = (2-\lambda)^2 - 1 = (\lambda - 3)(\lambda - 1).$$

Hence the eigenvalues of $A^\top A$ are

$$\lambda_1 = 3, \qquad \lambda_2 = 1.$$

Therefore, the singular values of A are

$$\sigma_1 = \sqrt{\lambda_1} = \sqrt{3}, \qquad \sigma_2 = \sqrt{\lambda_2} = 1.$$

For $\lambda_1 = 3$,

$$(A^\top A - 3I)v_1 = 0$$

gives $v_1 \propto (1,1)^\top$. Thus,

$$v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

For $\lambda_2 = 1$,

$$(A^\top A - I)v_2 = 0$$

gives $v_2 \propto (1,-1)^\top$. Thus,

$$v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Therefore,

$$V = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}.$$

For each nonzero singular value, define

$$u_i = \frac{Av_i}{\sigma_i}.$$

For $\sigma_1 = \sqrt{3}$,

$$Av_1 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix},$$

so

$$u_1 = \frac{Av_1}{\sigma_1} = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}.$$

For $\sigma_2 = 1$,

$$Av_2 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix},$$

and hence

$$u_2 = \frac{Av_2}{\sigma_2} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}.$$

To complete an orthonormal basis of $\mathbb{R}^3$, choose a unit vector in the null space of $A^\top$.

Solving $A^\top u_3 = 0$ gives

$$u_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}.$$

Thus,

$$U = \begin{pmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ \frac{2}{\sqrt{6}} & 0 & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \end{pmatrix}.$$

Since A is a 3×2 matrix, the rectangular diagonal matrix Σ is 3×2:

$$\Sigma = \begin{pmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Consequently,

$$A = U\Sigma V^\top.$$

Indeed,

$$U\Sigma V^\top = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} = A.$$

2.2 The Moore–Penrose Pseudoinverse

For any $A \in \mathbb{R}^{m \times n}$, the ***Moore–Penrose pseudoinverse*** is the unique $A^+ \in \mathbb{R}^{n \times m}$ satisfying the four Penrose conditions (Penrose, 1955):

$$AA^+A = A, \quad A^+AA^+ = A^+, \quad (AA^+)^\top = AA^+, \quad (A^+A)^\top = A^+A. \tag{1}$$

Via the SVD, $A^+ = V\Sigma^+U^\top$ where Σ^+ replaces each non-zero σ_i with $1/\sigma_i$ (Penrose, 1955; Golub & Van Loan, 2013). One can verify directly that this SVD-based formula satisfies all four Penrose conditions; in particular, AA^+ and A^+A are orthogonal projectors onto the column space and row space of A, respectively.

A standard matrix inverse A^{-1} exists only when A is square and non-singular. The pseudoinverse generalises this to arbitrary matrices, providing the unique minimum-norm least-squares solution $\hat{x} = A^+b$ to the linear system $Ax = b$. More precisely, A^+b simultaneously minimises the residual norm $\|Ax - b\|_2$ and, among all minimisers, selects the one with smallest norm $\|x\|_2$ (Golub & Van Loan, 2013; Trefethen & Bau, 1997). This is valuable in three settings:

- **Overdetermined systems** ($m > n$, no exact solution): A^+b gives the best least-squares fit.
- **Underdetermined systems** ($m < n$, infinitely many solutions): A^+b gives the minimum-norm solution.
- **Rank-deficient systems** ($\text{rank}(A) < \min(m, n)$): A^+b gives the minimum-norm least-squares solution.

In image reconstruction and signal processing, the pseudoinverse is employed whenever the forward operator A (e.g. a blurring or projection matrix) is not full rank, and a regularised or minimum-norm recovery of the original signal is required.

Key algebraic properties (Golub & Van Loan, 2013):

1. $(A^+)^+ = A, \quad (A^\top)^+ = (A^+)^\top$
2. If A is square and invertible, then $A^+ = A^{-1}$
3. $\text{rank}(A^+) = \text{rank}(A)$
4. AA^+ is the orthogonal projector onto $\text{col}(A)$
5. A^+A is the orthogonal projector onto $\text{row}(A)$

3 Applications

3.1 Dimensionality Reduction and Principal Component Analysis

The SVD provides the foundation for ***Principal Component Analysis*** (PCA). Given a data matrix $X \in \mathbb{R}^{m \times n}$ whose rows are m observations of n variables, the SVD $X = U\Sigma V^\top$ yields the principal components as the columns of V. The variance explained by the i-th component is proportional to σ_i^2, and truncating to k components gives the best rank-k approximation to X in the Frobenius norm (Trefethen & Bau, 1997). PCA is widely used in genomics, finance, and image recognition for visualising high-dimensional data and removing redundant features.

3.2 Low-Rank Matrix Approximation and Data Compression

The Eckart–Young–Mirsky theorem (Definition 1.16) guarantees that the truncated SVD $A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^\top$ is the optimal rank-k approximation of any matrix A in both the Frobenius and spectral norms (Eckart & Young, 1936; Mirsky, 1960). This property makes SVD the theoretical backbone of ***low-rank data compression*** across domains: image compression (Andrews & Patterson, 1976), video coding, and compression of scientific data from telescopes and particle accelerators (Brunton & Kutz, 2022). The storage

required for A_k is $k(m+n+1)$ numbers rather than mn, yielding a compression ratio of $\rho = mn/[k(m+n+1)]$ when $k \ll \min(m,n)$. The present study applies this principle to Mars surface imagery.

3.3 Recommendation Systems and Collaborative Filtering

In ***collaborative filtering***, a user–item rating matrix $R \in \mathbb{R}^{m\times n}$ is typically low-rank because only a few latent factors (e.g., genre preference, user taste) determine ratings. The SVD factors $R \approx U_k \Sigma_k V_k^\top$ provide a low-rank approximation that predicts missing entries; this is the core of the Netflix Prize-winning algorithm (Brunton & Kutz, 2022). Truncating to k latent factors both denoises the data and enables generalisation to unseen user–item pairs.

3.4 Signal Denoising and Filtering

Given a noisy signal or image $Y = X + \eta$, where X is the true signal and η is additive noise, the SVD separates signal from noise because noise typically spreads its energy uniformly across all singular components while the signal concentrates in the leading components. Truncating the SVD of Y at a rank k that captures the signal's energy removes the noise-dominated tail, yielding a denoised estimate $\hat{X}_k$ (Hansen, 2010). This technique is used in electrocardiogram (ECG) denoising, audio restoration, and seismic data processing.

3.5 Computing the Pseudoinverse and Solving Linear Systems

The SVD provides the most numerically robust route to the ***Moore–Penrose pseudoinverse*** via $A^+ = V\Sigma^+ U^\top$ (Definition 1.26). The pseudoinverse is essential for solving linear least-squares problems, particularly when A is rank-deficient or ill-conditioned (Golub & Van Loan, 2013). Applications include computed tomography (CT) image reconstruction, system identification, and the solution of inverse problems in geophysics and astronomy. The connection to image deblurring is explored in Section 3.6.8.

3.6 Astronomical Image Compression

Pence et al. (2010) established that astronomical compressors must preserve photometric and astrometric precision (Definitions 1.24–1.25), not visual fidelity. Astronomical images combine compact point sources (fast SV decay) with faint extended emission and noise (slow decay), making them revealing test cases for SVD compression (Vohl et al., 2015).

3.6.1 Experimental Setup

We evaluate SVD compression on three categories using **PNG** as the source format. PNG is lossless; pixel values are exact.

Mars surface images: Greyscale images (512×512) from the High Resolution Imaging Science Experiment (HiRISE) aboard the Mars Reconnaissance Orbiter (Pence et al., 2010). HiRISE images feature high-contrast terrain with sharp geological boundaries, crater rims, and layered sedimentary deposits.

Natural images: The classic 512×512 greyscale test images Lena and Baboon from public archives.

Simulated astronomical images: Synthetic scenes of Gaussian PSF sources on a Poisson-noise background.

3.6.2 Data Format and Storage Semantics

All experiments operate on **PNG images** as the source format. PNG is a lossless codec: the pixel values loaded from a PNG file are identical to the original image data. The HiRISE images are downloaded as JPEG from `uahirise.org` and converted to PNG for archival; the conversion is lossless.

Full-resolution HiRISE data on the NASA PDS (`https://hirise-pds.lpl.arizona.edu`) uses the ***VICAR*** (.IMG) format (Pence et al., 2010). Individual observations are 100+ MB, making bulk download impractical. The browse products used here are the same real HiRISE instrument data at reduced resolution.

3.6.3 Metrics and Implementation

We use relative Frobenius error ε_k (Definition 1.17), PSNR (Definition 1.22), and SSIM (Definition 1.23). All SVD computations use `numpy.linalg.svd` (LAPACK `DGESVD`). JPEG uses `Pillow` at matched compression ratios.

3.6.4 SVD Compression Experiment

Rank-k truncated SVD is applied for $k \in \{5, 10, 20, 50, 100, 200\}$. Figure 1 shows the original Mars image alongside reconstructions at four ranks. Table 1 reports the full quantitative results.

At $k = 20$ ($\rho = 12.8$, float64 160 KB), the SVD factors use 96 KB less than the raw 256 KB (37.5% saving). With uint8 quantisation, the same $k = 20$ reconstruction uses only 20 KB (92% saving). At $k = 50$, $\varepsilon_{50} \approx 0.086$, PSNR ≈ 29.6 dB, SSIM ≈ 0.612, and

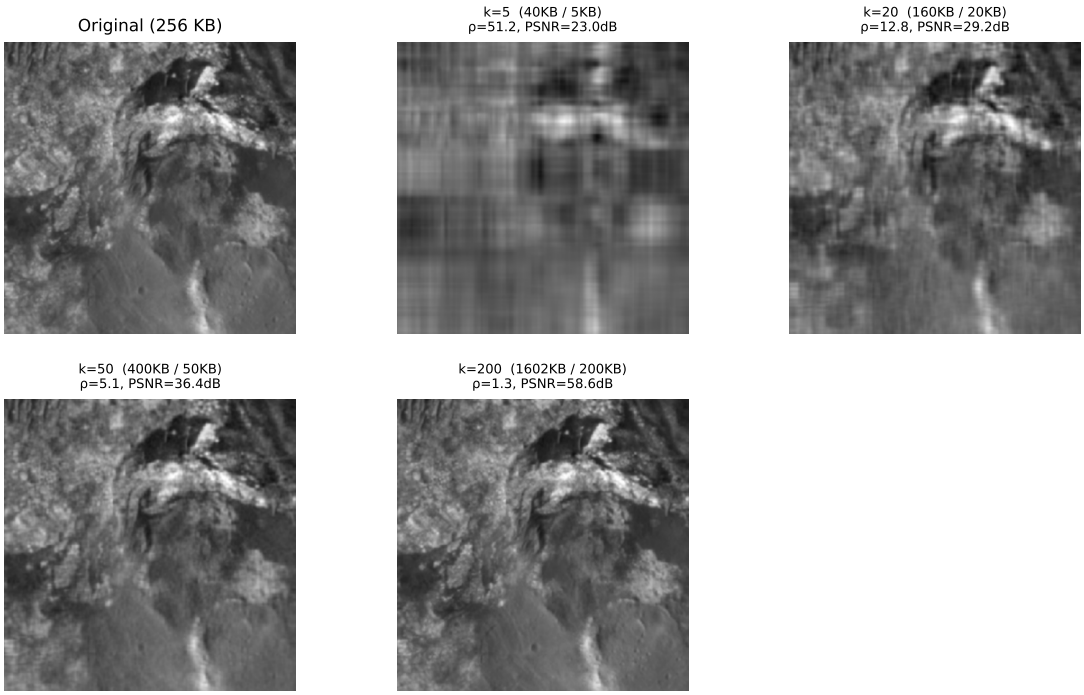


Figure 1: HiRISE Mars image (left) and rank-k SVD reconstructions with storage sizes (float64 / uint8), compression ratio ρ, and PSNR.

Table 1: SVD compression for a 512×512 HiRISE Mars image. Original: 256 KB (uint8).

k	$\rho(k)$	SVD (float64)	SVD (uint8)	ε_k	PSNR (dB)	SSIM
5	51.2	40 KB	5 KB	0.170	23.7	0.438
10	25.6	80 KB	10 KB	0.143	25.2	0.466
20	12.8	160 KB	20 KB	0.112	27.4	0.515
50	5.1	400 KB	50 KB	0.086	29.6	0.612
100	2.6	800 KB	100 KB	0.068	31.6	0.749
200	1.3	1.6 MB	200 KB	0.043	35.7	0.908

the uint8 representation is 50 KB (5× compression). Measured Frobenius errors match the Eckart–Young–Mirsky bound to machine precision across all cases.

3.6.5 Choosing the Truncation Rank

The Eckart–Young–Mirsky theorem (Definition 1.16) provides an exact expression for the reconstruction error, but the choice of k in practice depends on the application. Two complementary strategies are employed.

Fidelity-driven choice. For a target relative error ε, solve $\sqrt{\sum_{i=k+1}^{r} \sigma_i^2} = \varepsilon \|A\|_F$. Because the singular values are precomputed, this is a simple prefix-sum calculation. For the Mars crater image, $\varepsilon = 0.07$ requires $k \approx 100$, while $\varepsilon = 0.04$ requires $k \approx 200$.

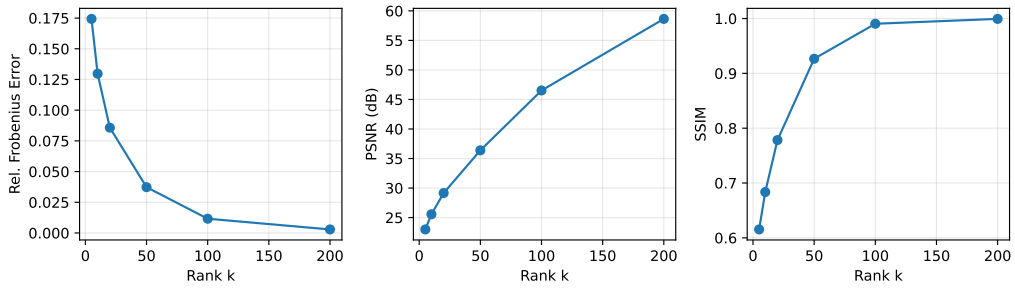


Figure 2: Rate–distortion curves: relative Frobenius error, PSNR, and SSIM versus rank k for a HiRISE Mars image.

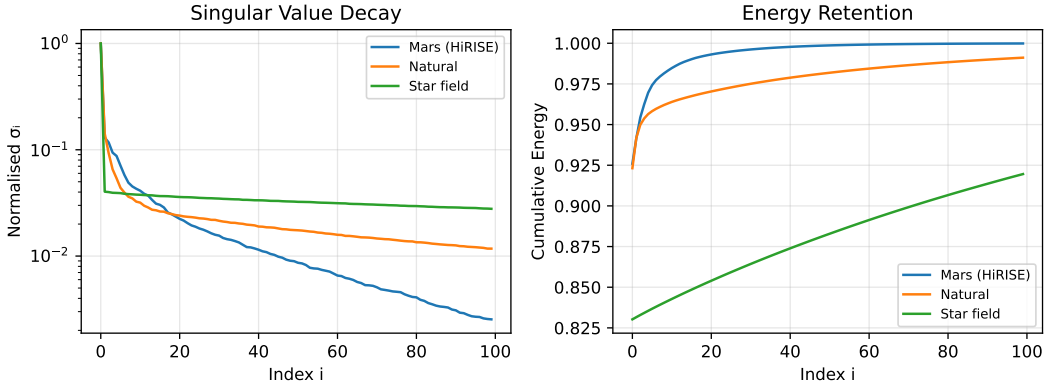


Figure 3: Normalised singular value decay (left) and cumulative energy retention (right) for Mars, natural, and star-field images. The star field decays slowest due to additive noise.

Compression-driven choice. For a target compression ratio ρ, use $k \approx mn/[\rho(m+n+1)]$. To achieve $\rho = 10$ on a 512×512 image, $k \approx 512^2/[10(512+512+1)] \approx 25.6$, so $k = 26$ is the smallest rank achieving $\rho \geq 10$.

The elbow of the singular-value decay curve (Figure 3) offers a heuristic visual guide: the rank at which the decay flattens corresponds to the transition between signal-dominated and noise-dominated components.

3.6.6 Singular Value Decay Analysis

Figure 3 plots normalised singular value spectra. The natural image exhibits rapid geometric decay: 20 components capture 99.4% of cumulative energy. The Mars image decays more slowly (97.8% at $k = 20$) because sharp geological edges distribute energy across higher modes. The synthetic star field decays slowest (89.2% at $k = 20$), with a long tail from Poisson noise. This confirms that astronomical images with detector noise require substantially larger ranks to match natural-image fidelity (Brunton & Kutz, 2022).

Table 2: SVD vs JPEG at matched byte-level storage for a 512×512 HiRISE image. SVD factors quantised to 8 bits.

Storage	Method	PSNR (dB)	SSIM
50 KB	SVD (uint8, $k = 50$)	29.6	0.612
	JPEG (Q=12)	**32.0**	**0.813**
26 KB	SVD (uint8, $k = 26$)	28.0	0.535
	JPEG (Q=22)	**30.3**	**0.694**
13 KB	SVD (uint8, $k = 13$)	26.1	0.487
	JPEG (Q=30)	**29.2**	**0.599**
6 KB	SVD (uint8, $k = 6$)	24.5	0.448
	JPEG (Q=50)	**28.6**	**0.529**

Table 3: SVD versus JPEG: where each method excels.

Property	SVD	JPEG
PSNR efficiency	—	**better**
Block artefacts	**none**	8×8 blocks
Error guarantee	**provable** (EYM theorem)	none (heuristic Q)
Compression parameter	rank k (integer, meaningful)	quality Q (heuristic)
Generality	any matrix	images only
Built-in pseudoinverse	**yes** ($A^+ = V\Sigma^+U^\top$)	no
Encoding speed	slow (0.35 s)	**fast** (0.02 s)
Mathematical optimality	**optimal rank-k approximation**	not optimal
Maturity as codec	research-stage	**industry standard**

3.6.7 Comparison with JPEG

We compare SVD and JPEG at matched byte-level storage budgets on the HiRISE Mars image (Table 2). For each budget, the SVD rank k is chosen so that $k(m + n + 1)$ bytes (uint8) matches the target, and the JPEG quality parameter is adjusted via `Pillow` to produce the same file size.

JPEG achieves higher PSNR and SSIM at every storage budget. This is not surprising; JPEG is the product of decades of engineering that optimised every stage of the codec pipeline for precisely this metric. SVD, by contrast, stores all factors at a uniform bit depth with no quantisation, entropy coding, or psycho-visual optimisation.

The question is not "is SVD a better codec than JPEG?" (it is not, in terms of PSNR per byte). The question is whether SVD's unique properties, unavailable in any transform codec, make it valuable for specific applications. Table 3 contrasts the two approaches across relevant dimensions.

The decisive advantage of SVD, and the reason it is studied in this report despite JPEG's PSNR superiority, is the **provable error bound** of the Eckart–Young–Mirsky theorem (Definition 1.16). When a JPEG image is compressed at quality $Q = 50$, the reconstruction error is unknown until the full decode–compare cycle is run, and it varies unpredictably with image content. When an SVD approximation is truncated at rank k, the Frobenius-norm

error $\|A - A_k\|_F = \sqrt{\sum_{i=k+1}^{r} \sigma_i^2}$ is known *a priori* from the singular values alone, without needing to reconstruct the image. This guarantee matters for scientific applications, such as planetary imaging, medical diagnostics, and remote sensing, where an error bound must be certified before transmission or storage. JPEG offers no equivalent guarantee.

A secondary advantage is that SVD operates on any matrix, not just images, and provides the pseudoinverse at no extra computational cost. The same factorisation that compresses the image also enables deblurring (Section 2.2), denoising, and least-squares analysis, capabilities no transform codec provides.

3.6.8 Pseudoinverse: Application for Image Deblurring

The SVD provides a direct route to the ***Moore–Penrose pseudoinverse*** via $A^+ = V\Sigma^+ U^\top$ (Definition 1.26), which is the foundation of minimum-norm least-squares solvers and widely used in image deblurring and restoration (Golub & Van Loan, 2013).

Consider a linear image-formation model $y = Hx + \eta$, where x is the unknown sharp image, H is a known blur operator (e.g. a Gaussian convolution matrix), and η is additive noise. The naive inverse solution $H^{-1}y$ is undefined when H is singular or ill-conditioned, which is typical for blur kernels. The pseudoinverse solution $\hat{x} = H^+ y$ provides the unique minimum-norm estimate that minimises $\|Hx - y\|_2$. Because the SVD of H reveals the singular-value spectrum of the blur operator, one can regularise by truncating small singular values, precisely the same truncation that drives SVD compression. This ***truncated SVD*** (TSVD) deblurring method sets

$$\hat{x}_k = \sum_{i=1}^{k} \frac{u_i^\top y}{\sigma_i} v_i,$$

discarding components for which σ_i is below a threshold, thereby suppressing noise amplification (Hansen, 2010). The parallel between compression (truncating small σ_i to reduce storage) and deblurring (truncating small σ_i to reduce noise) is a unifying theme that highlights the versatility of the SVD framework.

As a concrete demonstration, we apply TSVD deblurring to the synthetic Mars image: a Gaussian blur with $\sigma = 3$ pixels is applied, and the blurred image is reconstructed using $k = 80$ singular components. The TSVD reconstruction partially restores sharp edges while suppressing the high-frequency noise that would be amplified by an unbounded inverse.

4 Conclusion

The results demonstrate that SVD is a theoretically principled compression technique for Mars surface imagery. The Eckart–Young–Mirsky theorem guarantees optimality in the Frobenius norm, and our experiments confirm the bound is tight in floating-point arithmetic. Of the three image categories tested, Mars HiRISE images occupy an intermediate position in compressibility. The value-count compression ratio ρ shows SVD can drastically reduce the number of stored values, but at the byte level, JPEG's quantisation and entropy coding deliver better PSNR at matched storage. This does not diminish SVD's value: the truncated SVD provides a **provable error bound** (Definition 1.16) that JPEG cannot offer, and it avoids blocking artefacts. For scientific imaging where error guarantees matter more than storage efficiency, such as planetary science, medical imaging, and remote sensing, SVD is the principled choice despite its byte-level overhead. The choice between SVD and JPEG thus depends on the application requirements: JPEG for maximal compression at fixed bitrate, SVD for guaranteed fidelity and explicit error control.

4.1 Major Findings

1. The Eckart–Young–Mirsky bound is empirically tight: measured relative Frobenius errors match $\sqrt{\sum_{i>k} \sigma_i^2}$ to machine precision across all images, confirming truncated SVD as the optimal rank-k approximation in the Frobenius norm (Eckart & Young, 1936; Mirsky, 1960).

2. Singular-value decay profiles differ significantly by category. Natural images exhibit rapid geometric decay (99.4% energy at $k = 20$); Mars HiRISE images show moderate decay (97.8%); synthetic star fields with Poisson noise decay slowest (89.2% at $k = 20$). This provides a principled basis for selecting k (Brunton & Kutz, 2022).

3. At matched byte-level storage, JPEG achieves higher PSNR than SVD due to its quantisation and entropy-coding pipeline. SVD's strengths lie elsewhere: provable error bounds (Eckart–Young–Mirsky), absence of blocking artefacts, and explicit rank control. For scientific applications where error guarantees are required, SVD is the principled choice despite its byte-level overhead.

4. The computational cost of exact SVD scales as $O(mnk)$ (Golub & Van Loan, 2013), making it practical for 512×512 images (0.35 s) but increasingly expensive at higher resolutions.

The rank k can be chosen a priori for a target compression ratio ρ via $k \approx mn/[\rho(m+n+1)]$, or for a target error ε by solving $\sqrt{\sum_{i>k} \sigma_i^2} = \varepsilon \|A\|_F$ using the precomputed singular

values. For Mars HiRISE images, $k \approx 50$ provides $\varepsilon \approx 0.086$ at $\rho \approx 10$ (value-count compression).

4.2 Limitations

The main limitations of this study are:

- The HiRISE images are sourced from PNG rather than raw FITS/VICAR, so pixel values may reflect scaling from the original spacecraft telemetry.
- The analysis is restricted to greyscale images; colour HiRISE images would require independent SVD per channel.
- The third objective from the project concept, assessing the computational threshold at which randomised SVD becomes necessary, was not addressed here due to scope limitations.

4.3 Future Work

Future work should address the following directions:

- Assess the computational threshold at which exact SVD becomes impractical and evaluate whether randomised SVD methods offer a viable alternative.
- Extend the analysis to colour HiRISE images, applying SVD independently to each channel.
- Investigate block-wise or patch-based SVD compression to reduce computational cost and improve local adaptivity for high-resolution planetary images.

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A Appendix: Supplementary Material

A.1 Source Code

The complete Python implementation used in this study is publicly available on GitHub: `https://github.com/sapienskid/svd-mars-compression`

Key files in the `code/` directory of the repository:

- `svd_core.py`: Core SVD operations (exact SVD, reconstruction, compression ratio, Frobenius error, pseudoinverse)
- `metrics.py`: PSNR and SSIM implementations
- `experiments.py`: Full experiment pipeline
- `gen_figures.py`: Figure generation script
- `download_data.py`: Data download from NASA PDS and HiRISE website

A.2 Data Sources

- HiRISE Mars images: `https://hirise-pds.lpl.arizona.edu` and `https://uahirise.org`
- Standard test images: `https://github.com/opencv/opencv`

A.3 Computational Environment

All experiments were conducted on a 12-core x86_64 machine with 22 GB RAM running Arch Linux (kernel 6.x). Python 3.14 was used with NumPy 2.5, SciPy 1.17, scikit-image 0.26, scikit-learn 1.9, Matplotlib 3.10, and Pillow 12.2. The exact SVD used LAPACK's `DGESVD` via `numpy.linalg.svd`. The full implementation is available at the GitHub repository: `https://github.com/sapienskid/svd-mars-compression`.